

The Vanishing Aurora Diamond

AI Mystery Detective Team Workshop

At 8:20 PM, Northbridge Museum lost power. Four minutes later, the priceless Aurora Diamond was gone from a locked display case. The glass was unbroken. Four people had possible motives. One access record changed the investigation.

Investigator name

Team

Date

Participant edition followed by a sealed facilitator solution

All people, places and events in this case are fictional.

Investigation Briefing

Your task is to determine the most likely explanation for the disappearance of the Aurora Diamond. A good investigation does not simply name a suspect. It connects claims to evidence, separates facts from inference, considers alternatives and states uncertainty.

Mission

- Reconstruct the timeline.
- Evaluate every clue.
- Compare all suspects using the same criteria.
- Challenge the leading theory.
- Recommend the most likely solution and explain its uncertainty.

Rules of investigation

- Use only information contained in this case book.
- Do not invent evidence, witnesses or events.
- A motive is not proof.
- Access by a card does not automatically prove who held it.
- Identify contradictions and innocent explanations.
- The Chief report must include missing evidence and human review.

How to use the book

Stage	Team action	Output
1	Read the incident and timeline	Initial theory
2	Assign the five agent roles	Specialist reports
3	Share reports in order	Growing context
4	Review the Chief verdict	Human decision
5	Open the sealed solution	Compare reasoning

Case File 001

Incident report

At 8:00 PM, curator Dr. Mira Sen displayed the Aurora Diamond inside a locked glass case in the Grand Gallery of Northbridge Museum. At 8:20 PM, a power failure darkened the gallery for four minutes. Emergency lighting was limited. At 8:30 PM, staff discovered that the diamond was missing. The display glass remained intact and showed no sign of forced entry.

Known facts

Item	Recorded information
Object	The Aurora Diamond
Location	Locked display case, Grand Gallery
Last confirmed	8:00 PM display inspection
Power failure	8:20 PM to 8:24 PM
Discovery	8:30 PM
Damage	No broken glass or forced entry
Lock system	Electronic; battery-backed access recording

Central questions

- When was the diamond removed?
- How was the case opened without damage?
- Who had means, opportunity and access?
- Which clues are strong, weak or misleading?
- What evidence would confirm or disprove the leading theory?

Suspect Files

Suspect	Possible motive	Statement and access
Lena Ortiz	Recently denied a promotion	Says she restarted the basement generator from 8:19 to 8:26. Her card opened the basement at 8:20.
Theo Park	Wanted publicity for the museum event	Says he remained on stage. A camera shows him continuously from 8:15 to 8:29.
Arjun Vale	Large private debt	Says he worked in the archive. His card opened the archive at 8:12 and the display case at 8:23.
Sofia Reed	Wanted an exclusive news story	Says she interviewed lobby guests. Three guests remember speaking with her during the blackout.

Fair comparison checklist

Criterion	Lena	Theo	Arjun	Sofia
Motive				
Means				
Opportunity				
Access				
Alibi support				
Evidence against				
Evidence in favor				

Witness Statements

Dr. Mira Sen, curator

“I inspected the diamond and locked the case at 8:00 PM. The electronic system accepts authorized cards and continues recording during a power failure because it has a battery. I did not reopen the case.”

Lena Ortiz, facilities coordinator

“The generator alarm appeared shortly before the blackout. I went to the basement and worked there from approximately 8:19 until 8:26. I crossed the wet courtyard earlier while inspecting an exterior door.”

Theo Park, event host

“I remained on the stage during the programme. The audience and camera were facing me. I did not enter the Grand Gallery during the blackout.”

Arjun Vale, archivist

“I worked in the archive throughout the blackout. My access card remained in the jacket I had left inside the archive. I carried a catalogue folder when I left because I was returning documents.”

Sofia Reed, journalist

“I was interviewing visitors in the lobby when the lights failed. I continued recording comments on my phone. Several guests were standing with me.”

Three lobby guests, combined note

“Each guest independently remembers speaking with Sofia during part or all of the blackout. None reports seeing her enter the gallery.”

Investigator note: A statement is evidence, but it may be incomplete or mistaken. Look for independent support.

Evidence Register

ID	Source	Observation
A	Electronic lock specification	Valid-card access is recorded during a power failure because the lock has a battery.
B	Display-case access log	Arjun Vale's card opened the display case at 8:23 PM.
C	Arjun's statement	He says his card remained in his jacket inside the archive.
D	Camera image at 8:25 PM	Arjun leaves the archive carrying a flat catalogue folder. The image does not show its contents.
E	Fiber examination	Blue velvet fibers were found inside the folder. The display cushion is made from blue velvet.
F	Muddy shoeprint	A print near the case matches Lena's boot size. Records show she inspected the case after crossing a wet courtyard earlier that afternoon.
G	Insurance record	If the diamond remains missing, insurance pays the museum rather than any named suspect.

Evidence classification

ID	Fact or inference	Strength	Alternative explanation
A			
B			
C			
D			
E			
F			
G			

Timeline Reconstruction

Time	Recorded event	Reasoning boundary
8:00	Diamond displayed and case locked	Confirmed inspection
8:12	Arjun's card opens archive	Card use does not prove who held it
8:15–8:29	Theo appears continuously on stage camera	Strong alibi for this period
8:19–8:26	Lena says she works on generator	Partly supported by basement access
8:20–8:24	Power failure	Likely opportunity window
8:23	Arjun's card opens display case	Strong access evidence; identity uncertain
8:25	Arjun carries a flat folder	Could conceal an item; image is not proof
8:30	Diamond reported missing	Exact removal time remains inferred

Your reconstruction

Most important time window

Event that changes the case

Contradiction

Missing event or information

AI Detective Team Assignments

Agent	Responsibility	Required output
Detective	Organize the incident	Timeline, confirmed facts and open questions
Evidence	Examine clues A–G	Facts, inferences, strength and contradictions
Suspect	Compare all four people	Motive, means, opportunity, access and alibi
Skeptic	Challenge the leading theory	Assumptions, alternatives and missing evidence
Chief	Synthesize all reports	Likely solution, confidence, weakness and next step

Handoff order

Case file → Detective → Evidence → Suspect → Skeptic → Chief → Human review

Report template

Agent role

Observations

Evidence references

Interpretations

Uncertainties

Recommended next step

Human Review Worksheet

Initial conclusion

Most likely suspect

Strongest evidence

Weakest evidence

Alternative explanation

Confidence and reason

Quality review

Question	Yes	No	Notes
Does every factual claim come from the case?			
Were all suspects compared fairly?			
Did the team separate motive from proof?			
Did the Skeptic offer a credible alternative?			
Does confidence match the uncertainty?			
Is missing evidence clearly stated?			

Final human decision

Accept, revise or reject the AI verdict

Reason

Next evidence to collect

Investigation Challenge

Complete the original investigation first. Then modify one condition and run the team again. Compare what changes and what remains stable.

Recommended second run

Remove Evidence E, the blue velvet fibers. Predict whether Arjun remains the leading suspect and whether confidence should decrease.

Comparison	Original run	Modified run
Leading suspect		
Chief confidence		
Strongest clue		
Best alternative		
Missing evidence		
Human decision		

Optional variations

- Strengthen Lena’s generator alibi with a timestamped maintenance log.
- Add a witness who saw someone carrying Arjun’s jacket.
- Require the Chief to cite evidence letters for every claim.
- Run Detective, Evidence and Suspect independently before synthesis.
- Limit each report to 100 words and observe information loss.

Facilitator Only Solution

Stop here if you are a participant. Open this section only after the investigation and group discussion.

Most likely explanation

Arjun Vale is the most likely suspect. His access card opened the display case at 8:23 PM during the blackout. Two minutes later, a camera showed him leaving the archive with a flat folder, and blue velvet fibers matching the display cushion were found inside that folder. His statement that the card remained in his jacket conflicts with the access record.

Why this is not certainty

The access system proves that Arjun’s card was used, not that Arjun personally used it. The camera does not show the diamond inside the folder, and blue velvet fibers could require further laboratory comparison. A different person might theoretically have taken the card or transferred the fibers. The evidence supports a leading explanation, not a proven legal conclusion.

Evidence interpretation

Evidence	Facilitator interpretation
A and B	Strongly establish authorized access during the blackout.
C	Creates a contradiction but does not identify who held the card.
D	Supports opportunity to conceal or transport the diamond.
E	Connects the folder to material used in the display.
F	A deliberate-looking distraction with a plausible innocent explanation.
G	Weakens a broad insurance-motive theory for the named suspects.

Best next evidence

- Recover and inspect the access card for handling evidence.
- Examine archive and corridor footage before and after the blackout.
- Search the folder and archive under appropriate fictional-case procedures.
- Compare fibers scientifically rather than relying only on color.
- Verify whether anyone could access Arjun’s jacket or borrow his card.

Teaching conclusion

Reward evidence-grounded reasoning, not an exact confidence percentage. A strong answer should identify Arjun as the leading suspect while clearly explaining the remaining uncertainty and the need for human review.